GG-S-00800(DSA-DM)
19 February 1965

INTERIM FEDERAL SPECIFICATION

SUCTION APPARATUS, SURGICAL, AUTOMATIC, THERMOTIC, MOBILE, 110 VOLT, AC-DC

This interim Federal specification was developed by the Defense Medical Supply Center, Third Avenue and 29th Street, Brooklyn, New York 11232, based upon currently available technical information. It is recommended that Federal agencies use it in procurement and forward recommendations for changes to the preparing activity at the address shown above.

1. SCOPE

1.1 This specification covers an automatic, thermotic type, drainage and suction apparatus, capable of providing gentle, intermittent suction.

2. APPLICABLE SPECIFICATIONS, STANDARDS AND OTHER PUBLICATIONS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal form a part of this specification:

Federal Specifications:

QQ-C-320	Chromium Plating (Electrodeposited).
TT-E-491	Enamel; Gloss, Synthetic (for Metal and Wood Hospital Furniture).
TT-P-659	Primer-Surfacer; Synthetic, Tints and White (for Metal and Wood Surfaces).
ZZ-T-831	Tubing; Rubber.
PPP-B-585	Boxes, Wood, Wirebound.
PPP-B-601	Boxes, Wood, Cleated-Plywood.
PPP-B-621	Boxes, Wood, Nailed and Lock-Corner.
PPP-B-636	Box, Fiberboard.

(Activities outside the Federal Government may obtain copies of Federal Specifications and Standards as outlined under General Information in the Index of Federal Specifications and Standards and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U.S. Government Printing Office, Washington, D.C., 20402.

(Single copies of this specification and other product specifications required by activities outside the Federal Government for bidding purposes are available without charge at the General Services Administration Regional Offices in Boston, New York, Atlanta, Chicago, Kansas City, Mo., Dallas, Denver, San Francisco, Los Angeles, Seattle, and Washington, D.C.

(Federal Government activities may obtain copies of Federal Specifications and Standards and the Index of Federal Specifications and Standards from established distribution points in their agencies.)

FSC 6515

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Military Specifications:

MIL-L-10547 Liners, Case, and Sheet, Overwrap; Water-Vaporproof or Waterproof, Flexible.

Military Standards:

MIL-STD-105 Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 Marking for Shipment and Storage.

(Copies of Military Specifications and Standards required by suppliers in connection with specific procurement functions shall be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publication. The following document forms a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids, or request for proposal shall apply.

UNDERWRITERS' LABORATORIES, INC.

UL499 Standards for Safety-Electric Heating Appliances.

(Underwriters' Laboratories Standards are available from the Underwriters' Laboratories, Inc., Walt Whitman Road, Melville, L.I., New York 11749, 207 East Ohio Street, Chicago, Illinois 60611, and 1655 Scott Boulevard, Santa Clara, California 95050).

3. REQUIREMENTS

3.1 Fire and casualty hazards.

3.1.1 The contractor shall submit to the contracting agency proof, prior to delivery, that the equipment he will supply under this specification conforms to the following requirements of the Underwriters' Laboratories Standard UL 499:

Temperature (Normal operation).

Power input.

Switches.

Automatic controls (thermostat).

Such proof shall consist of a laboratory report or listing by the Underwriters' Laboratories, Inc. The label or listing of the Underwriters' Laboratories shall be acceptable as evidence that the equipment conforms to the above. A copy of each laboratory report shall be furnished to the contracting officer.

3.1.2 In lieu of the label or listing by the Underwriters' Laboratories Inc., the contractor may submit a laboratory report by a nationally recognized testing agency adequately equipped and competent to perform such services, as shall be evidenced by the attachment of its seal, stamp or label to the article, or the issuance of a certificate. The contractor shall submit the laboratory report which shall include test results and methods used to determine compliance, to the contracting officer for approval. The Government reserves the right to require a facilities report showing the qualifications of the testing laboratory and to conduct a facilities survey to determine its acceptability.

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3.1.3 Compliance with the above requirements regarding fire and casualty hazards, does not absolve the contractor from complete compliance with the requirements of this specification in order to secure the acceptance of his commodity.

3.2 Materials.

3.2.1 Metal. Metal shall be as specified herein and suitable for the intended purpose.

3.2.2 Glass. The glass for the bottles shall be clear, well annealed, and free of cracks, flaws, and other defects. Examination for strain shall be conducted as specified in 4.4.3.

3.3 Design. The suction apparatus shall consist of a stand, pump, control panel, and suction system.

3.3.1 Stand. The stand shall have four uprights, a top, and a shelf. A push handle shall be located on one side. The uprights shall be flared just below the level of the shelf to form legs and to give stability. A caster shall be affixed to the end of each leg. The uprights, handle, and shelf shall be made of standard low-carbon cold-rolled steel. The uprights shall be of 1 inch, tubular, welded or seamless construction, and shall be made of 0.0478 inch thick (No. 18, U.S. Standard Revised gage) steel. The handle shall be made of 3/8-inch steel rod. It shall be located on the left hand side when facing the stand and near the top. The handle shall span the depth of the stand and shall extend vertically from the side of the stand and be bent upward. The top of the handle shall not extend over the level of the top of the stand. The top shall be made of an acid-resistant, thermosetting, resin laminate plastic. The edge shall be trimmed with a T-type polished aluminum strip. The four casters shall be swivel type and shall have 2 inch rubber wheels.

3.3.2 Pump. The pump (heating chamber) shall be mounted directly under the top. It shall be equipped with a power cord fitted with a connector plug. The pump shall be nonmechanical, operating on the principle of expansion and contraction of air when subjected to variations in applied heats. It shall consist essentially of the following components:

(a) Pump chamber.
(b) Heating unit inside of chamber.
(c) Thermostat.
(d) Pilot lamp - 125 volts.
(e) 16 ohm, 20 watt resistor.
(f) 0.25 microfarad, 400 volt capacitor, 1.75 ohm resistance.
(g) Line switch.
(h) High-low switch for control of vacuum.
(i) Connecting power cord, rubber covered, 2 wire, with a connector plug having two parallel male contacts. The length of the cord shall be 10 feet.
(j) Chamber pressure outlet valve.
(k) Chamber suction inlet valve.
(l) Pump control panel mounted by means of either corrosion-resisting steel screws or brass screws with a chrome plating.

3.3.2.1 Control panel. A control panel shall be placed facing the front of the stand (on the 13 inch width), just below the top and in front of the pump. The control panel shall have an "on - off" switch, a neon pilot light to show when the pump is in operation, and a vacuum control switch. The switches and pilot light shall be centrally located.

3.3.3 Suction system. A length of tubing shall connect the suction inlet valve of the pump chamber and a pint capacity bottle suspended in a receptable just below the pump unit. The bottle shall, in turn, be connected to a larger gallon capacity bottle by means of tubing. A length of tubing

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which passes through a snubbing connection mounted on the front lower left edge of the panel shall extend from the larger bottle. A straight connector tip shall be affixed to the end of the tubing. A clip for holding the end of the tube when not in use shall be located on the right hand side of the stand. Tubing shall be connected by means of fittings which are inserted in holes in rubber stoppers placed in the mouths of the bottles. The large bottle shall rest on an overflow cutoff valve unit, designed to hold the bottle, which is mounted on the shelf. The small bottle shall act as a safety overflow trap in order to prevent damage to the thermotic pump. The larger bottle shall act as a receptacle for fluids and shall be marked to indicate the maximum point at which it must be emptied.

3.3.3.1 Bottles. The glass bottles shall be of a wide mouth type with cork finish. The vacuum receptable bottle shall have a capacity of 1 gallon. The overflow safety bottle shall have a capacity of 1 pint.

3.3.3.2 Tubing. The tubing shall conform to the applicable requirements of ZZ-T-831 for grade L, type I, heavy wall. The inside diameter shall be 3/16 inch and the wall thickness shall be 3/32 inch. The tubing extending from the vacuum receptable bottle to the straight connector tip shall be a minimum of 5 feet long.

3.3.3.3 Stoppers. The stoppers shall be made of rubber and shall be of sufficient size and resiliency to fit tightly in the mouth of each bottle. The perforations in the stoppers shall be of such size that the fittings will fit snugly.

3.3.3.4 Hold down clamps. Each bottle will be equipped with hold down clamps in order to prevent the stoppers from becoming loose. The hold down clamp for the vacuum receptable bottle shall be of such design as to permit it to be easily removed and replaced. It shall be made of corrosion-resisting steel wire.

3.3.3.5 Fittings. The fittings shall be either polished corrosion-resisting steel or brass with chrome plating.

3.4 Performance.

3.4.1 Pump. The thermotic pump shall operate on 110 volts, 60 cycles, alternating and direct currents. It shall be noiseless, positive in action, automatic, and operate continuously without attention.

3.4.2 Suction. Suction shall be produced which is capable of being adjusted, by a vacuum control switch, to a minimum of 90 millimeters (mm.) and to a minimum of 120 mm. of mercury. It shall be capable of continuing indefinitely when tested as specified in 4 4.2.

3.4.3 Overflow valve. When fluid reaches a predetermined weight in the gallon vacuum bottle, it shall activate an overflow valve which shall open the suction line, thus preventing floodings and damage to the thermotic pump. The pump shall be put back in operation by emptying the suction bottle.

3.4.4 Tube snubbing connection. The tube snubbing connection shall prevent the tube from pulling the vacuum receptacle bottle over when the tube is pulled.

3.5 Dielectric strength. There shall be no electrical breakdown when tested as specified in 4.4.1.

3.6 Measurements. The overall nominal measurements of the stand, excluding the handle, shall be 13 inches wide, 9-1/2 inches deep, and 33 inches high.

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3.7 Finish

3.7.1 Painted surfaces. All exposed surfaces of the uprights, shelf, and other surfaces not otherwise finished in chrome plating, polished corrosion-resisting steel, aluminum, or plastic laminate shall be painted.

3.7.1.1 Cleaning and preparation. Before finishing, parts to be painted shall be cleaned with suitable solvents and treated with phosphoric acid or other appropriate chemicals, to insure rust resistance of the metal and to provide a satisfactory bond for the finishing materials. The parts shall be putty glazed and baked. Castings shall be polished prior to surface treatment and application of primer.

3.7.1.2 Primer. After cleaning and preparation, the parts shall be given one coat of a synthetic primer conforming to TT-P-659. The primer shall be baked on for not less than 45 minutes at a temperature of 250° Fahrenheit (F.), or for an equivalent baking period at a higher or lower temperature, as recommended by the paint supplier and approved by the contracting officer. The primer shall be applied so as to level out and produce a uniform surface free from drips, runs, grit, seeds, streaks, blisters, fatty edges, nonuniformity of color, wrinkles, or other surface defects.

3.7.1.3 Enamel. After the primed parts have been allowed to cool, all exposed surfaces shall be painted with two coats of synthetic gloss enamel conforming to class B of TT-E-491. Each coat shall be baked on separately for not less than 45 minutes at a temperature of 250°F., or for an equivalent baking period at a higher or lower temperature, as recommended by the paint supplier and approved by the contracting officer. The enamel coats shall be applied so as to level out and produce a final, smooth, uniform surface free from runs, wrinkles, orangepeel, grit, seeds, blisters, fatty edges, nonuniformity or color, or other surface defects, and without areas of thin or no film. Color of the enamel shall be silvertone gray.

3.7.2 Plated surfaces. Surfaces of the fittings and handle, except those parts made of polished corrosion-resisting steel or aluminum, shall be chromium plated as specified in QQ-C-320 for class 1, type I.

3.7.3 Top. The plastic laminate covering the top shall be a neutral gray.

3.8 Markings.

3.8.1 Vacuum receptacle bottle. The vacuum receptacle bottle shall be marked with the word "VACUUM". It shall have graduations extending from a 700-milliliter (ml.) level to a 2600-ml. level in increments of 100 ml. At the 2600-ml. level it shall be marked as follows: "DO NOT FILL ABOVE THIS LINE." All markings shall be permanent.

3.8.2 Safety overflow bottle. The safety overflow bottle shall be permanently marked as follows: "CAUTION-FLUID MUST NOT ENTER THIS PUMP."

3.8.3 Control panel. The control panel shall be appropriately marked to indicate whether the switch is "on" or "off". The vacuum control switch shall show a "90 mm." setting and a "120 mm." setting. On the right lower corner of the panel the rated voltage, amperage, and type of current shall be indicated. All markings on the panel shall be permanently affixed thereon.

3.8.4 Identification marking. Each suction apparatus shall be permanently and legibly marked with the following information:

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Serial number.
Name or registered trademark of the manufacturer.
Contract or purchase order number.
Letters U.S.

Decalcomania or markings secured by adhesives shall not be used.

3.9 Service data. The contractor shall enclose, with each unit, two copies each of one or more booklets which contain complete instructions for installation, operation, maintenance, and a list of component parts. Wiring diagrams shall be included for electrical items or components. Each part of the unit other than common fasteners, etc., shall be identified in the service data and parts list by the manufacturer's part number. As soon as practicable after award of the contract or purchase order, the contractor shall furnish to the contracting officer, via the cognizant Government inspector, two preliminary copies of his proposed service data, for review and comment by the Defense Medical Supply Center. The preliminary service data shall be submitted sufficiently early to permit adequate review by the Defense Medical Supply Center (based on a maximum period of 30 days for such review) and to allow incorporation by the contractor of required revisions and corrections in the final service data without delaying delivery under the contract or purchase order. In addition, the contractor shall mail 15 copies of the approved service data to the contracting officer for distribution.

3.10 Workmanship. Workmanship shall be first class throughout. The suction apparatus shall be free from defects which detract from its appearance or may impair its serviceability.

4. SAMPLING, INSPECTION AND TEST PROCEDURES

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Inspection. Inspection, as used in this specification, is defined as both examination (such as visual, auditory, etc., investigation without the use of special laboratory appliances or procedures) and testing (determination by technical means of physical and chemical properties) of the item.

4.1.2 Certificates of quality. When available, certificates of quality, supplied by the manufacturer of the component or material, listing the specified test method and test results obtained, may be furnished in lieu of actual performance of such testing by the contractor.

4.2 Sampling.

4.2.1 For examination. Sampling for examination shall be conducted in accordance with MIL-STD-105.

4.2.2 For tests. Sampling for tests shall be conducted in accordance with MIL-STD-105.

Component (Unit of product)	Characteristic	Require-ments	Test procedures	Inspection level	AQL
Complete item	Dielectric strength	3.5	4.4.1	100%	
Suction system	Suction	3.4.2	4.4.2	100%	
Bottles	Annealing	3.2.2	4.4.3	S-3	

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*Acceptance number shall be zero regardless of lot size. A minimum of three samples shall be taken regardless of lot size.

4.3 Examination. The suction apparatus shall be examined to determine compliance with all requirements contained in this specification.

4.4 Tests. Tests shall be conducted to determine compliance with specification requirements.

4.4.1 Dielectric strength. The suction apparatus shall be subjected to a gradually increasing potential of twice the rated voltage plus 1000 volts which shall be maintained at this value for a period of one minute.

4.4.2 Suction. The capability of the suction apparatus to produce a vacuum as specified in 3.4.2 shall be tested by use of a mercury manometer.

4.4.3 Annealing. Annealing of glass for the bottles shall be determined by a polarscope. Samples showing more than a slight degree of ring strain or longitudinal strain shall be rejected.

5. PREPARATION FOR DELIVERY

5.1 Packaging.

5.1.1 Unit of issue. One (1) suction apparatus, as specified, constitutes one unit of issue.

5.1.2 Level A.

5.1.2.1 Unit packaging. Stand and control panel shall be adequately protected by cushioning with corrugated pads, cellulose wadding, or other suitable material so as to prevent marring, scratching or other damage. Pump and bottles shall be packaged separately in fiberboard cartons which shall contain sufficient cushioning material or fittings to insure against shifting or breakage. Cartons and wraps shall be secured to prevent accidental opening.

5.2 Packing.

5.2.1 Level A.

5.2.1.1 Exterior container. Each unit packaged as specified in 5.1.2.1 shall be overpacked in an exterior container which shall be designed for type 2 load, and constructed in accordance with PPP-B-585, style 3, for class 3 use; PPP-B-601, table II, using type III, class 1 plywood; PPP-B-621, table IV; or PPP-B-636, type CF, class weather-resistant. Component parts shall be secured within the exterior container to prevent shifting or damage.

5.2.1.2 Case liner. Each level A wood box shall be lined with a waterproof case liner conforming to MIL-L-10547. Closure and sealing shall conform to applicable paragraphs of appendix thereto. Case liner shall not be required for fiberboard boxes.

5.2.1.3 Closure. Closure of wood boxes shall be in accordance with appendix of applicable box specification. Closure of each fiberboard box shall be as specified in the appendix, and waterproofing shall conform with 30.4, of PPP-B-636.

5.2.1.4 Strapping. Strapping, when required, shall be in accordance with appendix of applicable box specification.

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4.1.2 Inspection. Inspection, as used in this

NOTE: Strapping will not be required for shipment forwarded to a receiving activity within the continental limits of the United States for storage and redistribution.

5.2.2 Level B.

5.2.2.1 Exterior container. Each unit shall be overpacked as specified in 5.2.1.1 except that exterior container shall conform to PPP-B-601, table I, using type III, class 1 plywood, or shall be designed for type 2 load, and constructed in accordance with PPP-B-585, style 3, for class 2 use; PPP-B-621, table I; or PPP-B-636, type CF, class domestic and method I closure shall be utilized. Component parts shall be secured within the exterior container to prevent shifting or damage.

5.3 Marking.

5.3.1 Interior packages. Containers shall be marked to identify contents.

5.3.2 Exterior container. Exterior container shall be marked in accordance with MIL-STD-129. In addition, the following legend shall be shown: "GLASS-HANDLE WITH CARE."

6. NOTES

6.1 Ordering data. Procurement documents should specify the following:

(a) Number and date of this specification
(b) Selection of applicable levels of packaging and packing (see 5.1 and 5.2).

6.2 This specification does not include all types, classes, grades, sizes, etc., of the commodity indicated by the title of this specification, or which are commercially available, but it intended to cover the types, etc., which are normally procured to meet Federal Government requirements.

Custodians:
Army - MD
Navy - MS
Air Force. - 03

Review activities:
Army. - MD
Navy. - MS
Air Force - 03

Preparing activity:
Defense Supply Agency. - DM

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on centers and each area shall then be drilled with holes spaced 1/4 inch

SPECIFICATION ANALYSIS SHEET

FORM APPROVED BUDGET BUREAU NO. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use of the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner and send to preparing activity.

SPECIFICATION INTERIM FEDERAL SPECIFICATION GG-S-00800(DSA-DM) SUCTION APPARATUS, SURGICAL AUTOMATIC, THERMOTIC, MOBILE, 110 Volt, AC-DC.

ORGANIZATION	CITY AND STATE

CONTRACT NO.	QUANTITY OF ITEMS PROCURED	DOLLAR AMOUNT
		$

MATERIAL PROCURED UNDER A ☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES" IN WHAT WAY?

4. REMARKS (ATTACH ANY PERTINENT DATA WHICH MAY BE OF USE IN IMPROVING THIS SPECIFICATION. IF THERE ARE ADDITIONAL PAPERS, ATTACH TO FORM AND PLACE BOTH IN AN ENVELOPE ADDRESSED TO PREPARING ACTIVITY)

SUBMITTED BY (PRINTED OR TYPED NAME AND ACTIVITY)	DATE

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CAUTION

NOTICE TO BIDDERS/OFFERORS

DO NOT CONDITION OR BASE YOUR BID/OFFER ON ANY CURRENT PROCUREMENT ON THE INFORMATION SUBMITTED ON THIS FORM SINCE ANY CHANGES OR DELETIONS IN THE SPECIFICATIONS MAY RENDER YOUR BID/OFFER NON-RESPONSIVE IN WHICH CASE IT CANNOT BE CONSIDERED FOR AWARD.

ld Under

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EFENSE MEDICAL SUPPLY CENTER
3RD AVENUE AND 29TH STREET
BROOKLYN, N.Y., 11232

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